

NIMBYism and the Housing Supply*

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Abstract

We develop a political economy model of the housing market in which the supply of new homes is shaped by a democratic vote among households. Voter support for additional housing is determined by the effect this vote has on households' expected discounted utility via changes in house prices. Households that own housing, and therefore typically experience financial losses from additional supply, generally oppose new housing: the “Not In My Backyard” (NIMBY) effect. Households that wish to buy a house and would benefit from cheaper housing support more supply. We find that support for additional supply is highest among households that are younger, low-income or are currently renting. We then simulate the impact of a temporary baby boom in the population. We find that those born during the baby boom attain higher rates of home ownership and greater net worth due to their greater political influence relative to other generations. However, this comes at the expense of higher house prices, lower home-ownership rates and less wealth for the subsequent generations. Calibrating the model to US demographic data, we find that this political economy mechanism can account for a 15 per cent increase in house prices since 1950. Using demographic projections, we project that population ageing will continue to place upward pressure on house prices into the 21st century.

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1 Introduction

The “Not In My Backyard” (NIMBY) effect describes the phenomenon of local residents opposing new housing developments in their neighbourhood. As the regulation of housing supply is often decentralised to local governments, residents can exert significant political power over any new developments within their area. The aggregate effect of NIMBYism can thus negatively impact the housing supply within an economy. In the US, for example, land is divided into “zones” which determine the size and use of new developments.¹ Only detached, single-family homes are allowed to be built in “R1” residential zones, even if high demand for multi-home housing (such as apartments) exists. Those who own homes within R1 residential areas have strong incentives to maintain the status quo at the expense of non-homeowners, who would benefit from multi-home housing in order to get on the housing ladder (Manville et al. 2020).

We develop a political economy model of the housing market in order to estimate the impact of NIMBYism and show how changing demographic trends can account for part of the recent rise in house prices. Specifically, we allow housing supply in a standard life cycle model to be determined by a democratic process at the local level: households vote according to their personal preferences and the final outcome is determined by a median-voter condition.

We then use this model to show that an ageing society exacerbates the effects of NIMBYism and implies a 15 per cent increase in house prices since 1950. Using demographic projections, we project that population ageing will continue to increase political pressure on the housing market in the 21st century. Finally we examine the dynamics of a temporary baby boom and analyse how it affects the level and distribution of wealth in the economy. We find that the generation born during the baby boom shifts the median voter over its life cycle, first toward additional construction and later toward restrictions on new supply. This increases its housing wealth and total net worth, but comes at the cost of decreased home-ownership rates and net worth for subsequent generations.

Our theoretical model follows the standard life cycle housing model as described by Iacoviello (2005). Over the life-cycle, an individual household typically rents housing until they have saved enough to meet their borrowing constraint and be able to put a deposit down on a house. Households also participate in the housing market by either upsizing (as their wealth grows or positive income shocks occur) or selling (if they receive a negative income shock or die). Instead of fixing the housing supply (as in Iacoviello 2005) or having it produced inelastically by a construction sector (as in Kaplan et al. 2020), we allow the supply of housing to be determined by a democratic vote of local households.

Household preferences over the supply of housing are formed according to the impact that a small change in price will have on their expected discounted utility. Households who gain from

¹Zoning laws were originally developed in the 1900s in order to improve “light and air” in residential areas - see Fischel (2004) for a historical overview. Land use regulation has also been used by white residents in the US to maintain and increase racial segregation (Sahn 2024; Trounstein 2020).

marginally higher prices will vote against new developments, whereas those who marginally lose out from higher house prices will vote in favour of new developments.² NIMBYist views in our environment are thus largely determined by households' ownership of housing and their expected probability of being a net-buyer in future periods. NIMBYist households are disproportionately older, wealthier, have higher incomes and are more likely to be homeowners, which is in line with the empirical literature surrounding NIMBYism (see Whitemore & BenDor (2018), Scally and Tighe (2015), and Hankinson (2018) for a range of surveys on the subject).

Our model primarily focuses on the financial incentives of NIMBYism, as households' own homes are the primary asset for most households.³ However there are several non-financial motivations that may lead a household to oppose new developments - ranging from infrastructure capacity concerns to simply disliking the proposed style of architecture or neighbourhood characteristics (Ahlfeldt 2011, Bosetti and Sims 2016). We allow for these non-financial preferences to affect households' votes but assume these preferences are stable over the life cycle. The financial incentive for NIMBYism, however, does vary over the life cycle and is analogous to an insider-outsider problem: outsiders (renters) would prefer more houses to be built until they become an insider (homeowner), and then seek to restrict supply.

As home-ownership rates and thus support for new developments change over the life cycle, the demographic make-up of an economy will play an important role in the aggregate effect of NIMBYism. During the mid 1940s, a baby boom occurred in a number of OECD countries including the US, UK, New Zealand, Australia, France and Canada (Greenwood et al. 2005). This baby boom lasted until the subsequent baby bust of the mid 1960s when birth rates returned to trend. The effect of the spike in the birth rate led to a generation known as the Baby Boomers which has taken up a significant share of the population over time. As those who are close in age will have similar preferences over housing on average (as a result of being at a similar stage of the life-cycle), those born during the baby boom will have outsized political power and thus have a greater influence over the aggregate housing stock. This is perhaps one reason why rents are increasing relative to median income over the long run as shown in Figure 1. It is also why the US home-ownership rate has been falling for younger people over time as shown in Figure 2.

²Households assume that increasing the housing supply decreases house prices within their local area. Although increasing housing supply may increase house prices via agglomeration effects, these are dispersed across a larger area (Duranton and Kerr, 2015) and so will be ignored by households voting within decentralised local government authorities that make up a greater metropolitan area. This dynamic underpins the NIMBY position: in principle, they may support increased housing supply - just not in their own backyard.

³Excluding the top 1% of the wealth distribution, home equity makes up 28.9% of total US household wealth and is comparable only to retirement accounts 32.8%. The next largest asset, stocks and mutual funds, makes up only 10.2% of total household wealth (see Eggleson et al. 2020).

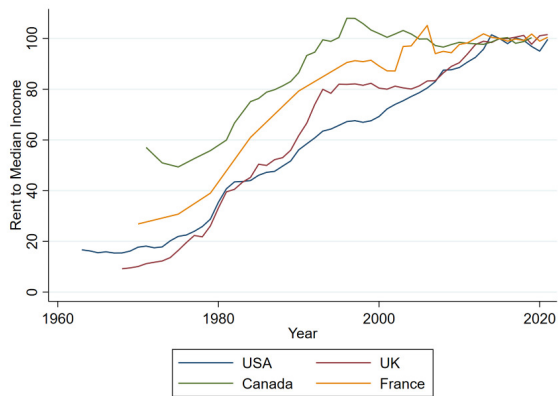


Figure 1: Rent to Median Income

Data source: CPS (2022).

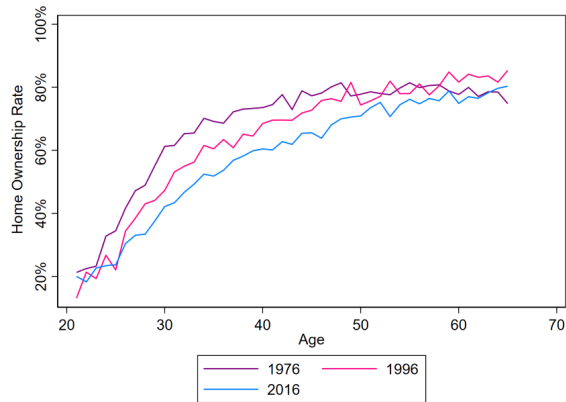


Figure 2: Home Ownership by Age Cohort

Data source: CPS (2022).

In this paper, we estimate how the changing demographic shares across ageing economies have changed the political-economic equilibrium in the housing market. As economies age, the balance of political power shifts in favour of NIMBYist households resulting in lower housing supply and higher house prices. We find that house prices increased by around 15 per cent due to these political-economic effects since the 1950s. Using demographic projections, as shown in figure 3, we find that the ageing economy is expected to keep the house price at or above its current level until 2100.

Finally, we use our political-economic model of the housing market to examine the distributional consequences of recent demographic shocks. We find that a temporary boom in the birth rate will initially decrease inter- and intra-generational wealth inequality as the boom generation initially supports lower house prices when they are disproportionately young renters. As the cohort ages, this effect reverses and the cohort exerts greater political pressure against the supply of new housing. This decrease in support for housing supply, and the resulting higher house price, leads to a persistent scarring effect on future generations who are unable to achieve home ownership at the same rate as previous generations, even after the initial baby boom generation has completely died off.

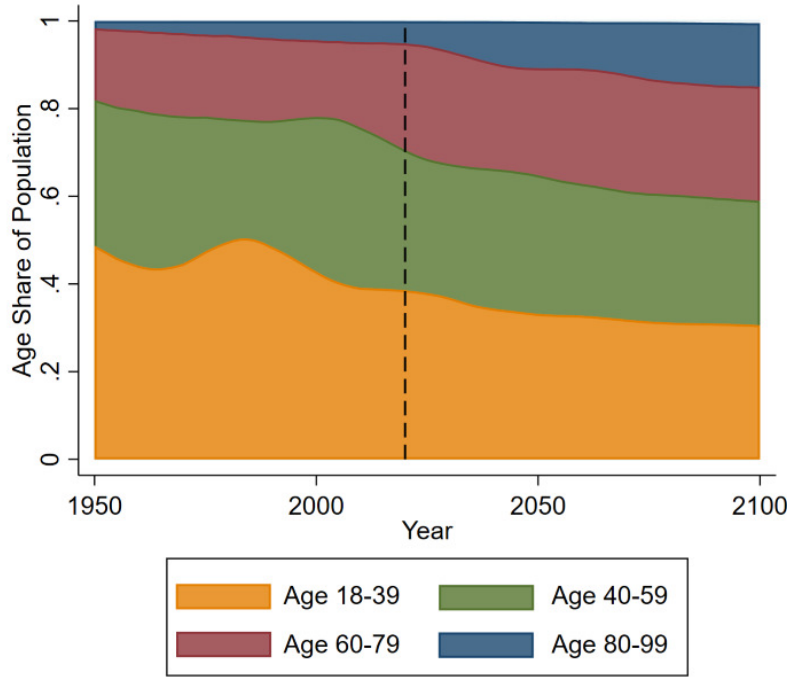


Figure 3: Age Share of Population Over Time

Data source: UN (2019). Note: dashed line indicates population forecasts from 2020 onward assuming medium immigration.

The remainder of the paper proceeds as follows. Section 2 discusses the related literature. Section 3 presents the model and voting mechanism. Section 4 describes the calibration. Section 5 reports the stationary and transition exercises. Section 6 presents robustness exercises, and Section 7 concludes.

2 Related Literature

This paper is related to the following strands of the literature. Firstly, there is growing literature surrounding the impact of regulation on the housing supply. The concept of NIMBYism is often attributed to Frieden (1979), who noted a rise in environmental regulations to impede new housing developments after the 1970s. Gyourko et al. (2013) support this view and argue that before the 1970s, housing construction tended to follow areas of high price growth. However, establishing a direct link between regulation and the housing supply is difficult due to the idiosyncratic forms that regulations can take in different localities (Gyourko and Molloy 2014). To get round this, Glaeser et al. (2006) offer indirect evidence of excess regulation by noting the existence of a relatively large wedge between prices and marginal construction costs. The difficulty in providing direct evidence stems from the endogeneity between land use regulation and the housing market. Recent papers have used boundary discontinuity design to address this

issue (Kahn et al. 2010, Turner et al. 2014, Kulka et al. 2022). For example, Anagol et al. (2021) exploit a zoning reform in Sao Paulo and show that housing developers apply for more permits in blocks with higher maximum densities. In their equilibrium model of housing supply this leads to a 2.2% increase in the total housing stock and 0.5% average price reduction. Similarly, Song (2021) constructs a nationwide data set of minimum lot area estimates and finds that doubling the US minimum lot area size would increase house prices by 14% and rents by 6%.

In theory, an increase in the supply of housing units may increase house prices via improved amenities and neighbourhood quality, or via agglomeration effects which increase economic activity (Herkenhoff et al. 2018, Chatterjee and Eyigungor 2017, Titman and Zhu 2024). In the immediate local vicinity, however, these demand effects do not appear to dominate the supply channel. For example, Li (2022) finds that within 500 ft, for every 10% increase in the housing stock, rents decrease by 1% and for every 10% increase in the condo stock, condo sales prices decrease by 0.9%. Similarly, Pennington (2021) finds that rents fall by about 2% for properties within 100 metres of a new housing project, and the effects decrease to zero when 1.5 km away. This illustrates the coordination problem in NIMBYism: the impact of extra housing within a homeowner’s immediate vicinity will negatively affect prices, whereas the benefits of agglomeration within the wider area can only occur if local residents accept extra housing in their back yard. As a result an emerging political economy literature has started to address this issue (see for example, Parkhomenko 2023, Bunten 2017, Duranton and Puga 2023 and Albouy et al., 2019). Using a difference-in-differences approach, Mast (2024) shows that when elections of town councils were localised from “at-large” to “ward” areas, housing permits decreased by 20%.

This paper is also related to an older strand of literature on political redistribution, such as Alesina and Rodrik (1994) and Persson and Tabellini (1994). The main idea of these papers is that when income and resources are distributed heterogeneously, the majority may vote for redistributive policies. These papers rely on a voting mechanism, the median-voter theorem, to determine what policy will be pursued. The theorem shows that when voting takes place over a single issue, and preferences are single-peaked, then the policy chosen under a democracy will be the one chosen by the median-voter.

Housing politics is a good candidate for the median-voter theorem as voting takes place at the local level and usually on specific propositions related to relaxing zoning regulations. The idea of homeowners as a specific voting class in local government decisions as a result of financial interests is related to the “homevoter hypothesis” (Fischel 1999; Fischel 2001). Einstein et al. (2019) examine planning meetings in Massachusetts and find that homeowners and older residents are overly representative in planning meetings. This is in line with voter turnout in local elections which in general tend to be from these two groups (Kogan et al. 2018, Oliver and Ha 2007). Similarly, Jiang (2018) finds that voter turnout increases for mayoral elections, relative to presidential elections, when home-ownership rates increase. In addition, Cheung and Meltzer (2013) find that the prevalence of Homeowners Associations is positively associated with

an increase in land use regulations. Similarly, Hilber and Robert-Nicoud (2013) find a positive correlation between the degree of physical development in an area and land use regulation. Both Ahlfeldt and Maennig (2015) as well as Dehring and Ward (2008) use public referendums to directly test the homevoter hypothesis. Ahlfeldt and Maennig (2015) use a referendum on the expansion of an airport in Berlin as a natural experiment and find greater support among homeowners as opposed to leaseholders. The support for the proposal comes from two effects: a consumption effect (through use of the amenity) as well as a wealth effect (via house price increases). Dehring and Ward (2008) use pre-vote changes in residential property values to test the probability of support for the new stadium for the Dallas Cowboys and find between a 0.9% and 1.2% increase in support for every \$1000 dollar increase in house prices.

Numerous surveys of voters have attempted to describe the demographics of voters who oppose new housing supply. Whittemore and BenDor (2018) find that older, less educated were more likely to oppose new developments, though they found no significant relationship between home ownership and NIMBYist views. By contrast Hankinson (2018) and Scally and Tighe (2015) found that home owners were significantly more likely to oppose new housing supply - particularly in their locality.

Finally, this paper is related to the literature surrounding life-cycle models with housing markets. Generally, life-cycle models assume exogenous house prices due to computational complexity (Fernandez-Villaverde and Krueger (2011), Iacoviello and Pavan (2013), Yang (2009)). There are, however, relatively few papers where aggregate shocks affect house prices (Justiniano et al. 2015). Our paper is most closely related to Ortalo-Magné and Prat (2014), who use a median-voter theorem in an overlapping-generations economy to show that increasing home-ownership rates decrease housing affordability. Similarly, Mankiw and Weil (1989) predict the growth rate of housing demand via changes in the age distribution alone and more recently Mumtaz and Šustek (2023) find similar results except housing demand peaks in the 2000s.

3 The Model

We outline a life-cycle model with housing and debt choices. The model features heterogeneous, life-cycle households that optimise non-durable consumption, housing and saving in liquid assets. A number of factors influence households' political support for additional housing supply. These include households' income and wealth, portfolio allocation, housing tenure and age over the life cycle. As demographics change, the composition of the voting population changes, and this alters political support for new construction. We use the model to study how an exogenous demographic shock changes voting over housing supply and, through this political channel, equilibrium house prices.

Households make housing and saving decisions in an environment in which future housing-supply restrictions are chosen politically. A demographic shock changes the future age and

tenure composition of voters. When a larger cohort is young, it raises the relative weight of renters and prospective buyers, who benefit from additional construction. As the same cohort ages into home ownership, political support shifts toward tighter restrictions on new supply. These changes in voter composition affect future supply restrictions and the market-clearing house prices associated with permitted supply.

Households are forward-looking, so the dynamic problem requires expectations about future housing-market conditions. These expectations enter through continuation values, the value of reaching future periods with a given state. We specify expectations using forecast rules that differ in the information households use about how future demographic change affects voting behaviour and, through the political process, future restrictions on housing supply. One forecast rule uses the current housing-market environment to summarise future conditions: households do not explicitly forecast the future demographic path or the political outcomes it will generate. The alternative forecast rules use the demographic path and the political housing-supply mechanism to forecast future voter composition, supply restrictions and prices. Comparing these forecast rules isolates how expectations about future demographics and politics shape capitalisation, housing choices and voting incentives.

A consumer of age j enters the economy at age 25 and lives for J periods, indexed by $j = 0, \dots, J - 1$, and maximises lifetime expected utility which consists of consumption c_t , housing services, h_t and a terminal bequest $v(w_J)$. Housing services depend on whether they own or rent the property. Individuals also have access to a liquid asset that pays an interest r_{t+1} . The household's lifetime utility function is given by

$$\sum_{j=0}^{J-1} \beta^j u(c_j, h_j) + \beta^J v(w_J) \quad (1)$$

where households optimise over consumption, c_t , liquid assets, a_t , and housing, h_t at each age j . The flow utility is increasing and strictly concave in c and h . Households receive utility from wealth they bequeath w_J . We assume the functional form

$$u(c_t, h_t) = \frac{(c_t^{1-\zeta} h_t^\zeta)^{1-\gamma}}{1-\gamma} \quad (2)$$

where the consumption of housing stock consists of either the amount of housing owned or rented

$$h_t = \begin{cases} \theta^{rent} h_t^{rent} & \text{if } h_t^{own} = 0 \\ h_t^{own} & \text{if } h_t^{own} > 0. \end{cases} \quad (3)$$

The curvature parameter γ determines households' risk aversion and inter-temporal elasticity of substitution and ζ denotes housing's share of total consumption. Rental housing provides a stream of utility that is discounted by θ^{rent} compared to self-owned housing. Finally there is a

warm glow bequest function when households die that provides utility

$$v(w) = \psi \frac{(w + \bar{w})^{1-\gamma}}{1-\gamma} \quad (4)$$

which generates a bequest function with decreasing marginal utility with respect to wealth, w , governed by the parameters $\bar{w}$ and ψ . Households supply one unit of labour inelastically and are paid a wage $y_{t,j}$. The wage they receive varies according to their productivity which is driven by the exogenous process

$$\begin{aligned} \log y_{i,j,t} &= g_j + z_{i,t} + \varepsilon_{i,t} \text{ for } j < J^{retired}, \\ z_{i,t+1} &= \rho z_{i,t} + \eta_{i,t+1}. \end{aligned} \quad (5)$$

The deterministic component g_j is the age profile. The persistent component $z_{i,t}$ and transitory component $\varepsilon_{i,t}$ are approximated with a finite-state Markov process calibrated to yield realistic income inequality, following Kindermann and Krueger (2022). The numerical process also includes two top-income states that capture “superstar” income realizations. Once workers reach retirement age $J^{retired}$, their retirement income is fixed as a function of income at retirement and is then carried as a retirement-income state. The income process is normalised to have a mean of 1, such that the price of housing p_t^h reflects the house price to income ratio.

Households may save or borrow liquid assets subject to a borrowing constraint where each household can borrow against a proportion, m , of the total value of the stock of housing they own

$$a_t \geq -m h_t^{own} p_t^h. \quad (6)$$

Net asset positions earn the risk-free rate when non-negative and are financed at the mortgage rate when negative. Let

$$r_t(a_t) = \begin{cases} r_t^a & \text{if } a_t \geq 0, \\ r_t^m & \text{if } a_t < 0, \end{cases} \quad (7)$$

where $r_t^m = r_t^a + \chi$ and χ is the mortgage spread. Households may buy or sell housing at price, p_t^h , subject to a minimum block size $\bar{h}$. Thus owned housing satisfies $h_t^{own} \in \{0\} \cup [\bar{h}, \infty)$. All property sales, including those that are sold as part of a bequest are subject to a transaction cost F_{sell} proportional to the total value of property sold, which is reflected by the indicator function I_{sell} . If they do not own any housing they must rent it from real estate firms at price p_t^{rent} . The household’s utility function is thus maximised subject to the general budget constraint⁴

⁴Noting that households only interact with the rental market if they own no housing stock of their own.

$$a_{t+1} = (1 + r_t(a_t)) a_t + y_t - c_t - p_t^{rent} h_t^{rent} - \left(1 - I_t^{sell} F^{sell}\right) p_t^h \Delta h_t^{own} \quad (8)$$

where $I_t^{sell} = \mathbf{1}\{\Delta h_t^{own} < 0\}$ is an indicator for households that choose to sell part or all of their current stock of housing.

New households enter with initial liquid wealth drawn from the calibrated entrant-wealth distribution. The mean of this distribution, a_{25} , is tied to average terminal bequests in the stationary economy.

3.1 Household's Problem

Each household's state vector is $\mathbf{s}_t = \{a_t, h_{t-1}^{own}, z_t\}$, where a_t is the stock of liquid assets, h_{t-1}^{own} is the amount of housing they owned at the end of period $t-1$ and z_t is the persistent component of their income. After retirement, the fixed retirement-income state described above is also carried forward, but we suppress it in the notation. The price of housing, $p_t \equiv p_t^h$, is an aggregate state variable. Expectations enter the household problem through the future price path used in continuation values. Forecast rules are indexed by λ . At date t , households evaluate housing and saving decisions using the perceived continuation price path

$$\hat{\mathbf{p}}_t(\lambda) = (p_t, \hat{p}_{t+1|t}(\lambda), \dots, \hat{p}_{T|t}(\lambda)), \quad (9)$$

where $\hat{p}_{t+s|t}(\lambda)$ is the house price that households use for date $t+s$. In the numerical transition path, the full perceived object is the collection of these date-specific continuation paths. The transition value function is therefore $V_{j,t}^\lambda(\mathbf{s}_t; \hat{\mathbf{p}}_t(\lambda))$. In the Bellman equations below, $\hat{p}_{t+1|t}(\lambda)$ denotes the relevant remaining continuation path from date $t+1$ onward, written compactly to keep the notation readable. When $\lambda = 0$, the forecast path collapses to the current-price rule:

$$\hat{p}_{t+s|t}(0) = p_t \quad \text{for all } s \geq 1. \quad (10)$$

In this case, households do not separately forecast future demographic changes or the political housing-supply restrictions those changes induce. Future house prices are therefore assumed to equal the current house price at date t . For larger values of λ , household forecasts place more weight on the demographic-political path implied by the model.

The value function for the household is given by the maximum of the three possible housing tenure options: rent (either because they owned no housing or sell it all that period), or are owners who choose to adjust their stock of housing or maintain their current level of housing. Households who choose to rent (and thus sell whatever stock of housing they owned at the beginning of the period) are able to freely adjust the amount of housing they rent each period. The value function for a household that choose to rent housing (and thus sell any previously owned housing stock) at age j is

$$\begin{aligned}
V_j^{rent,\lambda}(\mathbf{s}_t; p_t) &= \max_{c_t, a_{t+1}, h_t^{rent}} u(c_t, \theta^{rent} h_t^{rent}) + \beta E_t V_{j+1}^\lambda(\mathbf{s}_{t+1}; \hat{p}_{t+1|t}(\lambda)) \\
&\quad s.t \\
a_{t+1} &= (1 + r_t(a_t)) a_t + y_t - c_t - p_t^{rent} h_t^{rent} + (1 - F^{sell}) p_t^h h_{t-1}^{own} \\
a_{t+1} &\geq 0 \\
h_t^{own} &= 0
\end{aligned} \tag{11}$$

The value function for home owners who choose not to adjust their (non-zero) stock of housing in period t is

$$\begin{aligned}
V_j^{non-adjust,\lambda}(\mathbf{s}_t; p_t) &= \max_{c_t, a_{t+1}} u(c_t, h_t^{own}) + \beta E_t V_{j+1}^\lambda(\mathbf{s}_{t+1}; \hat{p}_{t+1|t}(\lambda)) \\
&\quad s.t \\
a_{t+1} &= (1 + r_t(a_t)) a_t + y_t - c_t \\
a_{t+1} &\geq -m p_t^h h_t^{own} \\
h_t^{own} &= h_{t-1}^{own}
\end{aligned} \tag{12}$$

finally, the value function for households who choose to buy or sell housing in period t is

$$\begin{aligned}
V_j^{adjust,\lambda}(\mathbf{s}_t; p_t) &= \max_{c_t, a_{t+1}, h_t^{own}} u(c_t, h_t^{own}) + \beta E_t V_{j+1}^\lambda(\mathbf{s}_{t+1}; \hat{p}_{t+1|t}(\lambda)) \\
&\quad s.t \\
a_{t+1} &= (1 + r_t(a_t)) a_t + y_t - c_t - (1 - I^{sell} F^{sell}) p_t^h \Delta h_t^{own} \\
a_{t+1} &\geq -m p_t^h h_t^{own} \\
h_t^{own} &= h_{t-1}^{own} + \Delta h_t^{own}
\end{aligned} \tag{13}$$

The final value function is the maximum of each of the three choices over housing tenure

$$V_j^\lambda(\mathbf{s}_t; p_t) = \max \left\{ V_j^{rent,\lambda}(\mathbf{s}_t; p_t), V_j^{adjust,\lambda}(\mathbf{s}_t; p_t), V_j^{non-adjust,\lambda}(\mathbf{s}_t; p_t) \right\} \tag{14}$$

The expected value function for the final year of life is the same for all types of housing tenure and is given by the warm glow bequest function above, where total wealth is the sum of liquid and illiquid wealth given the current housing price.

$$w = a + (1 - F^{sell}) p_t^h h^{own} \tag{15}$$

3.2 Real Estate Firms

Real estate firms borrow from households by issuing debt in the form of the liquid asset to fund the purchase of housing, which is rented out to households that do not own their own home. Real estate firms operate in a perfectly competitive market, and thus the rental price is pinned down by the cost of borrowing, a fixed cost of providing rental services, the price of purchasing house stock and any expected capital appreciation driven by the drift in the price of housing. We assume that real estate firms are able to borrow against the full value of their housing stock. The zero profit condition pins down the price of rental stock as the function of the cost of housing, capital gains from housing and the interest rate

$$p_t^{rent} = \phi^{rent} + r_t^a p_t^h - \left(\hat{p}_{t+1|t}(\lambda) - p_t^h \right), \quad (16)$$

where ϕ^{rent} is the per unit cost of providing rental services. The final term is the expected capital gain from holding housing, evaluated using the same one-period price forecast that enters household continuation values. Rental firms break even under this perceived price path; ex post profits can differ from zero when perceived and realised price changes differ. In the numerical exercises we restrict attention to paths for which the implied rental price remains positive.

3.3 Housing Supply

The aggregate economy consists of a unit measure of neighbourhoods across which all types of households are equally distributed. Aggregate demographic changes are therefore interpreted as common changes in the demographic composition of representative local housing markets. This abstracts from spatial sorting and from differences in the local concentration of particular demographic groups. The neighbourhood's housing supply in period $t + 1$ is determined in each neighbourhood by a simple majoritarian voting mechanism in which households vote over the amount of permitted housing in an area. Only households who currently reside in that neighbourhood can vote to determine the next period's level of new housing. Once the quantity of housing permits has been determined by the households' collective vote within the neighbourhood, the supply of housing is costlessly produced by a perfectly competitive construction sector in $t + 1$. This is similar to Kaplan et al. (2020), who model a perfectly competitive construction sector in which all rents from land permits accrue to the government. New construction is physically costless, but permits are scarce. House prices therefore reflect the scarcity value of permitted housing. Permit rents accrue to the government or absentee landowners and are not rebated to households.

An increase in the housing supply will reduce the price of housing in the locality and households vote understanding that an increase in housing permits will lead to a small persistent decrease in the cost of housing. An increase in the housing supply will also affect the price of rental properties by lowering the input price for the real estate firms. As a result, each household

in a neighbourhood will vote to increase (or decrease) the availability of housing permits in their locality based on how the induced change in housing costs affects their expected welfare. Given the quantity of permitted housing, the house price adjusts to clear the local housing market. We therefore represent the welfare effect of a marginal supply restriction by the fixed proportional price perturbation ϵ^p that such a restriction implies. In the transition calculations, the perturbation is applied to the perceived price path that represents this persistent price effect. Let $\widehat{\mathbf{p}}_t^{+\epsilon}(\lambda)$ denote the path that raises the current and future prices relevant for this marginal restriction by the factor $1 + \epsilon^p$, and define

$$\Delta V_{j,t}^\lambda(\mathbf{s}_t) = V_{j,t}^\lambda(\mathbf{s}_t; \widehat{\mathbf{p}}_t^{+\epsilon}(\lambda)) - V_{j,t}^\lambda(\mathbf{s}_t; \widehat{\mathbf{p}}_t(\lambda)). \quad (17)$$

In the stationary problem this collapses to the corresponding fixed perturbation of the stationary house price. Households vote to restrict housing supply when

$$\Delta V_{j,t}^\lambda(\mathbf{s}_t) + \sigma \geq 0. \quad (18)$$

The perturbation size ϵ^p is fixed throughout the computations, so σ is calibrated in the same utility units as this finite-difference welfare comparison. The parameter σ represents a non-financial preference over the supply of housing. A positive value of σ shifts households toward restricting new supply, while a negative value shifts them toward allowing additional construction. For example, households may view new housing negatively as a result of congestion effects on infrastructure, or positively because of the new amenities that greater density could bring. Households may also hold ideological views about housing supply, which this parameter would capture as well.

In principle these preferences could vary by household, either due to changes in age, productivity, or as a result of an idiosyncratic shock. However, allowing for such heterogeneity would require introducing a larger set of additional parameters that would then need to be calibrated. To limit our degrees of freedom, we therefore calibrate these preferences as a simple, fixed constant across all households regardless of age, income or wealth. As a result, this parameter is effectively the average non-financial preference for all households. We test this assumption in Section 6 and find that the main results are little changed under alternative assumptions.

The impact of a small perturbation to the price of housing will vary depending on household's ownership of liquid assets, housing and their current level of productivity. If a small increase in the price of housing increases their expected lifetime utility, then this expression will be positive and they will vote to restrict housing permits.

3.4 The Median Voter Theorem

Applying the median-voter logic, equilibrium in the housing market occurs when the median voter is indifferent to a marginal change in housing permits, evaluated through the price change

that permits imply.⁵ Let $\mu_{j,t}(ds)$ denote the distribution of individual states conditional on age j , and let $\omega_{j,t}$ denote the population weight on that age. For a perceived price path $\hat{\mathbf{p}}_t(\lambda)$, the age-weighted vote share for restricting housing supply is

$$S_t(\hat{\mathbf{p}}_t(\lambda)) = \sum_{j=0}^{J-1} \omega_{j,t} \int \Theta_{j,t}(\mathbf{s}; \hat{\mathbf{p}}_t(\lambda)) \mu_{j,t}(ds), \quad (19)$$

where

$$\Theta_{j,t}(\mathbf{s}; \hat{\mathbf{p}}_t(\lambda)) = \mathbf{1}_{\{\Delta V_{j,t}^\lambda(\mathbf{s}) + \sigma \geq 0\}}. \quad (20)$$

In the stationary accounting exercise, the perceived path is constant at p^h , and the local political equilibrium chooses the house price such that

$$S_t(p^h) = \frac{1}{2}. \quad (21)$$

At this price, one half of the electorate supports a marginal restriction on new permits and one half does not. We interpret the resulting price as the market-clearing house price conditional on the permit scarcity implied by the political outcome. Along a transition, the same vote-share object is applied period by period to generate the price path.

The welfare effect of a marginal supply restriction is determined by the household's current portfolio, tenure status and expected future housing demand. Figure 4 shows how these household characteristics correlate with the decision to restrict supply. Renters and households that expect to buy additional housing tend to support additional permits because new supply lowers purchase and rental costs. These households tend to be younger, have lower incomes and hold less wealth. By contrast, homeowners who are unlikely to buy additional housing tend to support restrictions because lower supply preserves the resale value of their housing wealth and the size of their bequest.

The induced equilibrium price of housing therefore varies with the relative size of these groups. An exogenous increase in young households, who disproportionately rent and expect to buy housing in the future, increases political support for additional supply. The house price then falls until the median household is indifferent over a marginal change in housing permits.

In order to abstract from strategic voting across dynasties, we assume different children and parents live in different neighbourhoods and thus do not affect the other's housing-market dynamics.⁶ Consequently even if an individual may benefit from a larger inheritance if the price of their parents home increases, it will not affect their vote on the optimal level of housing supply

⁵Strictly speaking, the standard median-voter theorem assumes single-peaked preferences over a one-dimensional policy space. In the quantitative exercise, the relevant object is the median-voter indifference condition for a marginal change in permits: at the equilibrium, one half of households support a marginal restriction on new permits and one half do not.

⁶According to Fu (2019) 61.5% adults live more than 10 miles away from their mother.

in their local neighbourhood.⁷

We also assume that households are unable to form coalitions when voting, nor bargain political support across time periods and vote solely based on their individual interest at time t .⁸

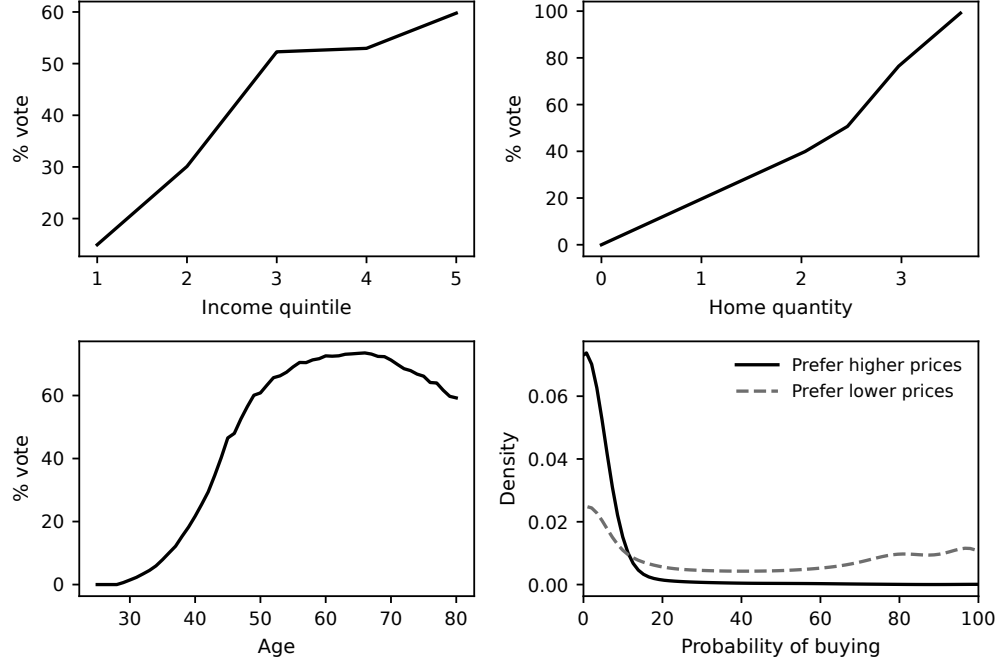


Figure 4: Household Support for Restricting Housing Supply

3.5 Stationary Equilibrium

For ease of notation, define the age distribution as $\Omega = \{\omega_{25}, \dots, \omega_{80}\}$, where ω_i is the share of the population at age i .⁹ For each exogenous age distribution Ω , we solve a stationary political-economic exercise: we hold age weights fixed and ask what house price satisfies the median-voter condition. The age distribution is therefore taken from the data or projection, rather than generated inside the model. The stationary accounting equilibrium consists of value functions, decision rules over consumption, $c(\mathbf{s}; p^h(\Omega))$, housing demand, $\{h^{rent}(\mathbf{s}; p^h(\Omega)), h(\mathbf{s}; p^h(\Omega))\}$, liquid savings $a(\mathbf{s}; p^h(\Omega))$, and votes over housing supply, $\{\Theta_j(\mathbf{s}; p^h(\Omega))\}_{j=0}^{J-1}$, which define the

⁷This assumption also makes our model robust to other forms of altruistic giving. For example, in addition to receiving a warm glow from giving to their descendants at death, households may care about their descendant's welfare directly and thus may wish to sell housing in order to give their children funds to purchase housing earlier. As families live in different neighbourhoods, however, this would still not change their voting behaviour. This is because parents with housing would still wish to restrict housing supply in their neighbourhood in order to receive the highest return on their house so they can maximise their child's welfare.

⁸This assumption allows us to abstract from any potential employment benefits derived from additional demand for the construction sector.

⁹We drop the t subscript for simplicity.

households' optimal allocations as a function of their individual state variables and the aggregate price of housing, and the prices of owned and rental housing, $p^h(\Omega)$ and $p^{rent}(\Omega)$. Within each accounting exercise, household formation is chosen only to hold the imposed demographic structure fixed. The transition calculations retain the same household decision problem, replacing the stationary house price with the forecast price path defined above. The age-specific distributions $\{\mu_j(ds; p^h(\Omega))\}_{j=0}^{J-1}$ are defined by the deterministic age process, the income process and households' optimal choices in response to these shocks. The full numerical computational method is outlined in detail in the Appendix.

3.6 Transition Equilibrium and Forecast Rules

For a given demographic path and forecast rule indexed by λ , we compute a transition path as follows. At each date, households optimise using a perceived continuation path for house prices. Their choices determine voter composition, voting determines housing-supply restrictions and the generated price path is the sequence of market-clearing prices conditional on the permitted housing stock. Let $\Omega = \{\Omega_t\}_{t=0}^T$ denote the demographic path and let $\widehat{\mathcal{P}}^{(n)} = \{\widehat{\mathbf{p}}_t^{(n)}\}_{t=0}^T$ denote the collection of date-specific continuation price paths used after forecast revision n . Given Ω and $\widehat{\mathcal{P}}^{(n)}$, we solve household decisions backward and simulate the economy forward, applying the political housing-supply rule at each date. The resulting voter composition and supply restrictions imply a generated house-price path $\mathcal{T}(\widehat{\mathcal{P}}^{(n)}; \Omega)$. Let $\mathcal{T}_t(\widehat{\mathcal{P}}^{(n)}; \Omega)$ denote the continuation segment of that generated price path beginning at date t . The bounded-learning specifications start from the current-price rule, $\widehat{\mathcal{P}}^{(0)} = \widehat{\mathcal{P}}^C$, and move each perceived continuation path part of the way toward the path generated by future demographics, voting outcomes and market clearing conditional on permitted supply:

$$\log \widehat{\mathbf{p}}_t^{(n+1)} = (1 - \lambda) \log \widehat{\mathbf{p}}_t^{(n)} + \lambda \log \mathcal{T}_t(\widehat{\mathcal{P}}^{(n)}; \Omega), \quad t = 0, \dots, T, \quad (22)$$

where n indexes belief revisions rather than calendar time. The parameter $\lambda \in [0, 1]$ indexes the weight placed in each revision on the path generated by future demographics, voting outcomes and market clearing conditional on permitted supply. When $\lambda = 0$, equation 22 leaves perceived future prices at the current-price rule. Positive values of λ move the perceived path toward the path generated by demographics, voting outcomes, supply restrictions and market clearing conditional on permitted supply. These paths are finite-revision expectation-sensitivity exercises: they vary how much of the demographic-political path enters household forecasts rather than selecting a single expectations concept. We do not interpret λ as a numerical relaxation parameter that is iterated to a common fixed point. If the update were iterated fully to convergence, positive values of λ would target the same fixed-point object. The reported exercises instead use finite forecast revisions and compare the resulting subjective forecast paths.

The parameter λ has a direct numerical interpretation. A forecast revision means applying

equation 22 once to the whole future price path. If the generated path were held fixed, the cumulative weight placed on that path after k revisions would be $1 - (1 - \lambda)^k$. This gives a useful scale for the plotted values: after ten revisions the corresponding weight is about 40 per cent for $\lambda = 5$ per cent, 65 per cent for $\lambda = 10$ per cent and 89 per cent for $\lambda = 20$ per cent.

A full rational-expectations transition would instead impose the fixed point

$$\hat{\mathbf{p}}_t = \mathcal{T}_t(\hat{\mathcal{P}}; \boldsymbol{\Omega}) \quad \text{for all } t, \quad (23)$$

so that households' perceived price path coincides with the path generated by their own housing choices, voting outcomes, supply restrictions and market clearing conditional on permitted supply. This is a strong informational requirement in the present environment: households would need to forecast not only future prices, but also how demographic change affects voter composition, political restrictions on supply, housing choices and the market-clearing prices implied by permitted supply. Our reported exercises therefore focus on how outcomes vary with the information households use when forecasting future prices.¹⁰

4 Calibration

We calibrate the model to capture salient features of the US housing market in the early 2000s prior to the housing boom and bust. The model period is one year. Households are economically active from ages 25 through 80, giving $J = 56$ annual periods. They work for 41 periods, through age 65, and receive the terminal bequest payoff at the end of life. The risk aversion parameter is set to 2, as is standard in the macroeconomics literature. The income process consists of the parameters for the deterministic age-profile, the persistent AR(1) component, and the transitory i.i.d. component of income. We estimate the parameters of the deterministic and stochastic income processes using data from the Panel Study of Income Dynamics.

The retirement income replacement rate is set at 50% of the final period's non-transitory income, based on Díaz and Luengo-Prado (2008). The risk-free interest rate is set at 2%, while the mortgage interest rate is 5%. The maximum loan-to-value (LTV) is set at 0.9 in line with Greenwald (2018). The spread between the saving rate and the mortgage rate is set at 3% matching the 30 year average difference between the two.

¹⁰This interpretation is close to Moll (2026), who argues that rational expectations over equilibrium prices in heterogeneous-agent models place a strong forecasting burden on agents and motivates restricted forecasting rules in which agents forecast prices more directly. We implement this idea through bounded-learning forecast rules for future house prices, with λ controlling how much information about the future demographic and political path enters those forecasts.

Table 1: Parameter Values

Parameter	Value	Interpretation	Source
J	56	Periods of life	KMV (2020)
$J^{retired}$	41	Periods of working life	KMV (2020)
r^a	0.02	Risk free rate	30 year average
r^m	0.05	Mortgage interest rate	30 year average
χ	0.03	Mortgage spread	30 year average
β	0.978	Discount factor	Internally Calibrated
γ	2	Risk aversion	KMV (2020)
ζ	0.75	Housing weight in utility	Internally Calibrated
F^{sell}	0.07	Cost of selling house	KMV (2020)
m	0.9	Maximum LTV	Greenwald (2018)
ψ	10	Bequest scaling	Internally Calibrated
$\bar{w}$	150	Bequest parameter	Internally Calibrated
σ	-2.4	Pro-supply Non-financial preference	Internally Calibrated, shifts 15% of the voters in steady state
ϕ^{rent}	0.05	Rental per unit cost	Internally Calibrated
θ^{rent}	0.9	Rental discount	Mian et al (2017)

We set the internally calibrated parameters $\{\beta, \sigma, \zeta, \bar{w}, \psi, \phi^{rent}\}$ using simulated method moments to match the following moments. The rental discount θ^{rent} is taken from the external calibration reported in Table 1.

Table 2: Calibration

	Model	Data
Share of owners with a mortgage	0.55	0.57
Share of renters	0.38	0.38
House price to Income Ratio	6.0	6.0
Home-ownership rate 25-35	0.38	0.40
Home-ownership rate 45-55	0.72	0.70
Home-ownership rate 65-75	0.71	0.75

The price point at which the median voter is indifferent to additional housing supply will be affected by σ , which captures households' non-financial preferences over the supply of housing. We calibrate σ such that the housing market is in equilibrium in 2000. Our calibration requires a non-financial preference in favour of approving more homes as homeowners almost always make up a large majority of the voting population.

5 Quantitative Analysis

5.1 Steady State

First we analyse how changes in the age distribution affect the housing market in a set of stationary exercises. For each year, we take the observed US age distribution as given and solve for the political-economic equilibrium implied by the model. We calculate the political-economic

equilibrium using demographic data from the US over the period 1950-2020. During this time there was substantial variation in the demographic structure, driven by the generation born in the post-war spike and how it reverberated over time, as shown in figure 3 at the beginning of the paper. For example, the mean age starts at 30 in 1950 and eventually falls as those born during the baby boom increase the share of young people in the labour and housing market. The mean age in the labour market troughs at 28 in 1970 as the baby boom subsides. The mean age then continues to steadily rise to 39 in 2020 as those born during the baby boom age.

To estimate the effect of the changing demographic composition for each year in this period, we solve for the equilibrium price that would result in the median voter being indifferent over an increase or decrease in housing supply. All other parameters are kept constant with the only change coming via variation in demographic structure. Because each calculation is stationary, the forecast-rule distinction does not materially enter this exercise; it becomes substantive in the transition analysis, where voter composition and political support for housing-supply restrictions change over time. Operationally, for each target age distribution we set household formation so that the demographic structure is constant in the absence of further shocks, and then solve for the associated stationary political-economic equilibrium house price.

Since home-ownership rates increase with age, we find that when the median population age increases the aggregate home-ownership rate rises. This increases political support for a reduction in the housing supply which increases house prices. House prices rise until there is enough political support for additional housing supply (via an increase in households who rent) such that the median voter is again indifferent to a small change in house prices.

Figure 5 shows the model’s political-economic equilibrium cost of housing over the post-war period alongside data on real rents.¹¹ This equilibrium cost represents the effects of demographic shocks on the voter coalition in the housing market, and thus ignores other factors such as changes in the business cycle, technology and financial crises. Accordingly one should not expect it to closely track actual costs that are affected by these non-demographic factors. Despite this, the equilibrium results fit the post-war trend in real rents quite closely.¹² In the model the post-war spike in the birth rate leads to a significant fall in the equilibrium house price starting in the 1970s as these households, who are mostly young renters, enter the housing market. As these households age and become homeowners they shift towards opposing the construction of new housing to protect the value of their stock of housing. Real rents show a similar trend peaking slightly earlier in the 1950s, before trending downwards for the next three decades reaching a low point in the 1980s then increasing steadily up until 2020.

¹¹In our model’s environment real rents move in line with the cost of housing, while abstracting from the impact of interest rates which will impact on the empirical price of housing.

¹²We use rents instead of house prices as a measure of housing costs to avoid the measure being distorted by the secular decline in interest rates over this period. Real rents are calculated using the average rent of a primary residence across US cities divided by headline CPI.

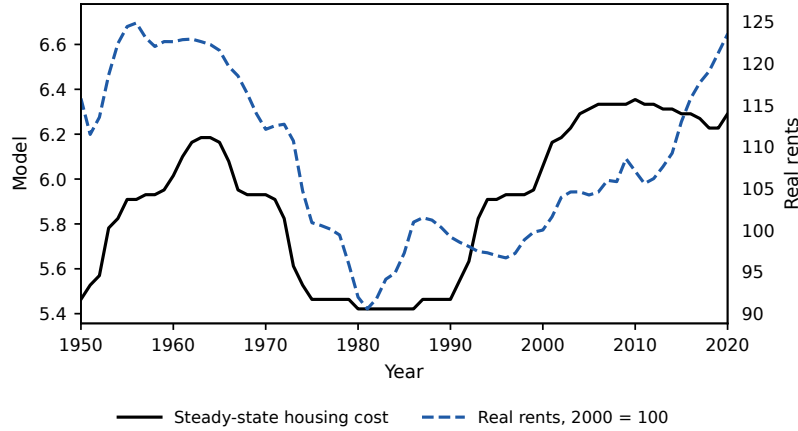


Figure 5: Stationary Accounting House Price and Real Rents

Using demographic projections out to 2100 we can estimate the impact the ongoing ageing of the US economy will have on the political economy of the US housing market as shown in figure 6. The projection series report representative-locality demographic paths for the US population under low, medium and high immigration assumptions. In the low and medium cases, ageing raises the share of voters who favour tighter supply at the initial price, so the stationary accounting price rises. Under high immigration, younger inflows partly offset ageing at first, but those cohorts later age into the homeowner coalition.

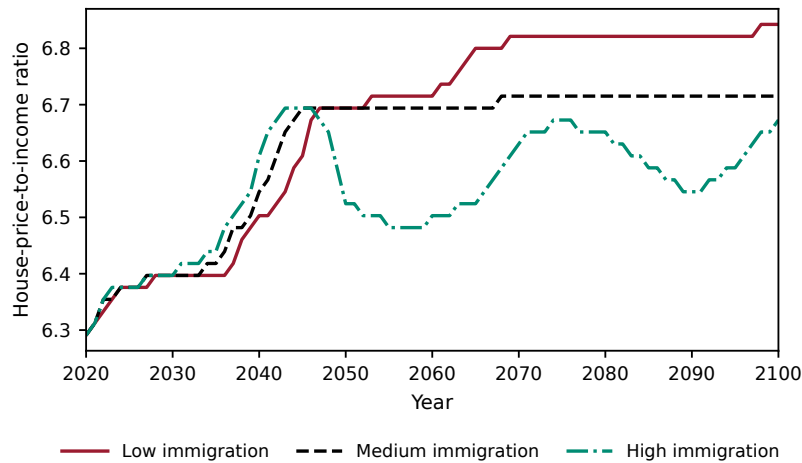


Figure 6: Stationary Accounting House Price Projections

5.2 Impulse Response Functions

We next calculate the dynamics of a demographic shock to the housing market in the form of a temporary baby boom. We start from the initial equilibrium and then increase the birth rate of

new households by 10 per cent for 10 years. After 25 years, when those who were born during the boom enter the labour and housing market, the effect of the spike in the birth rate starts to affect the demographic structure of households in the model. We then calculate forecast-rule transition paths across the subsequent 80 years to trace the effect of the baby boom through time.

Figure 7 reports the perceived house-price paths in the temporary baby-boom experiment, measured as log deviations from the initial price. We plot $\lambda = 0, 5, 10$ and 20 per cent. When $\lambda = 0$, households do not use demographic-political forecasts. Instead, future housing conditions in continuation values are summarised by the current house price at date t . As λ increases, perceived future prices place more weight on the path implied by future demographics, voting outcomes, supply restrictions and market clearing conditional on permitted supply.

The shape of the figure follows the life-cycle timing of the shock. Perceived prices are nearly flat at first, before rising as the boom generation reaches housing-market age and changes the relative size of the cohorts whose housing choices and votes determine supply restrictions. The price path then falls back because the demographic shock is temporary and the pressure passes through the age distribution. Higher values of λ dampen the hump because households' forecasts incorporate more of this future demographic and political adjustment, rather than extrapolating only from current market conditions.

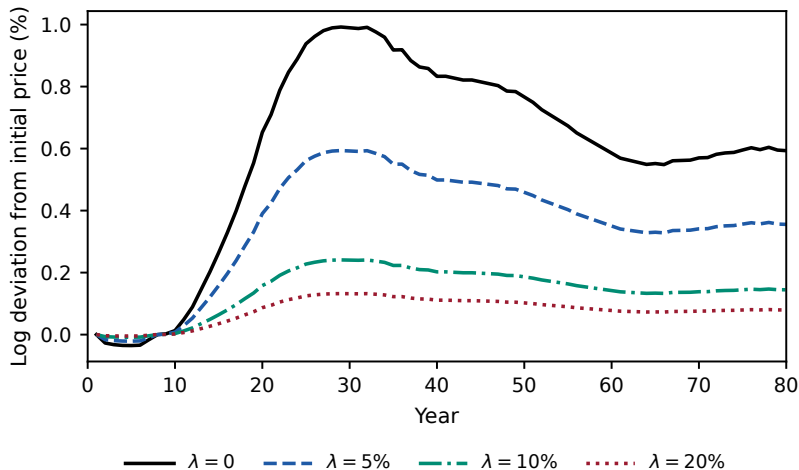


Figure 7: Forecast Price Paths for a Temporary Birthrate Increase

Figure 8 reports net worth and consumption for the generation born during the temporary baby boom. Each series is measured relative to households of the same age in the initial stationary economy, so a value above one means that the boom cohort has higher net worth or consumption than same-age households before the shock. As in Figure 7, higher values of λ dampen the house-price response to the demographic shock, which compresses the net-worth swing for the boom cohort. Consumption follows a smoother life-cycle profile, but it first dips below and then

risks above the age-specific initial value as housing and saving decisions adjust over time.

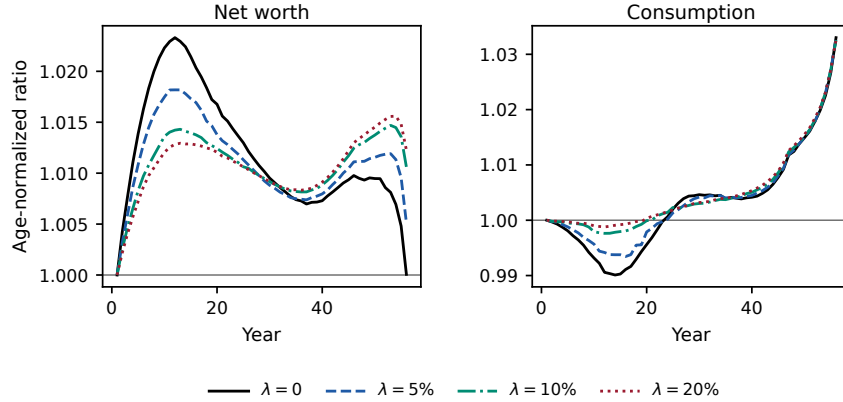


Figure 8: Boom-Cohort Effects of a Temporary Birthrate Increase

6 Robustness and Extensions

We report two sensitivity checks in the main text. The first varies the non-financial preference term that shifts the voting threshold in the temporary baby-boom transition. The second changes the weights assigned to voters in the stationary historical exercise. Both checks point to the same interpretation: changes in voter age and tenure composition drive the results. The magnitude varies with the political calibration and voting weights, but the demographic timing of the mechanism is unchanged.

6.1 Non-Financial Preference Calibration

Figure 9 varies the non-financial preference term in the voting rule. This term shifts the level of support for restricting housing supply and therefore changes the stationary house-price-to-income ratio. The transition paths are similar. The initial fall, later rise as the boom cohort enters the housing market, and return toward the stationary allocation change little because the experiment is driven mainly by voter-composition changes over time. Changing the non-financial preference term shifts the average level of political support, but it has little effect on the marginal political force created by the larger cohort.

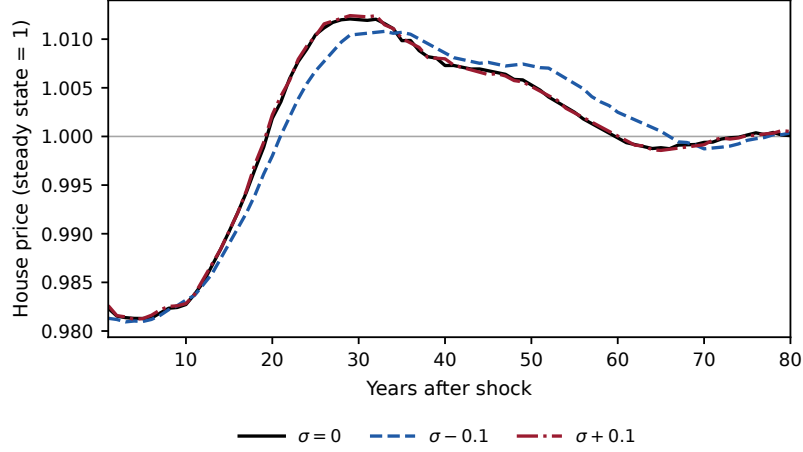


Figure 9: Robustness to Non-Financial Preference Calibration

6.2 Alternative Voting Weights

Figure 10 changes how household votes are aggregated in the stationary historical exercise. We compare the one-person-one-vote specification with voting weights based on age-specific turnout and log household net worth. Turnout weighting has little effect on the implied house-price path because it changes the political weight of the age groups driving the demographic mechanism only slightly. Log-net-worth weighting has a larger quantitative effect, since political influence shifts toward older and wealthier homeowners as the economy ages. Even in this case, however, the historical increase remains positive, although smaller.

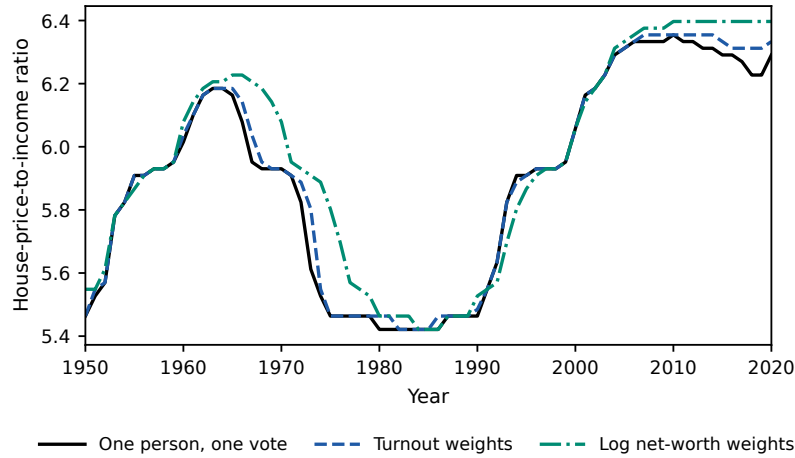


Figure 10: Alternative Voting Weights in the Stationary Historical Exercise

Table 3 summarises the same historical accounting exercise across the voting weights used in figure 10, together with an additional unlogged net-worth case. Holding the house price fixed at its year-2000 model value, the historical age distribution raises support for restricting supply

by 8.2 percentage points between 1950 and 2020 under one-person-one-vote. The stationary accounting-equilibrium price rises by 14.3 per cent over the same period. Weighting votes by turnout gives a similar change in political support and a 12.9 per cent price increase. Weighting votes by household net worth lowers the implied price change, but still produces a 7–8 per cent increase. The historical result is therefore not driven only by the one-person-one-vote specification, although the voting weights affect the magnitude of the channel.

Table 3: Historical Accounting and Voting Weights

Specification	Support for restricting supply		Equilibrium price
	1950	2020	1950–2020 change
One person, one vote	40.2	48.4	14.3%
Turnout weighted	43.0	51.3	12.9%
Log net-worth weighted	45.1	52.3	8.4%
Net-worth weighted	64.7	69.3	7.1%

Notes: The support columns report the share of weighted votes in favour of restricting housing supply, evaluated at the model’s year-2000 house price. The final column reports the percentage change in the stationary accounting-equilibrium house price between 1950 and 2020 under the same voting rule. All calculations use the historical age distributions and hold the remaining model parameters fixed.

7 Conclusion

Local planning regulations and the political-economic constraints that bind them have long been defining features of housing markets across high-income countries. Because housing is a large component of homeowners’ financial portfolios, some households have a direct financial incentive to protect this value by voting in favour of regulations that restrict the supply of new housing. In this paper, we show how post-war demographic changes shifted relative generation sizes and, consequently, political support for additional housing supply. In the model, this demographic-political mechanism implies a trough-to-peak increase in the house-price-to-income ratio of roughly 15 per cent. Furthermore, future demographic projections imply continued upward pressure on house prices over the coming decades.

Unpacking the distributional impacts of this generational shift, we find that a one-off increase in the birth rate changes the median voter over the life cycle. When the cohort is young, political support shifts toward additional construction; when it ages into home ownership, political support shifts toward restrictions on new supply. The transition exercises show that this mechanism is sensitive to how households form expectations about future demographic and political conditions. Under the current-price rule, households do not separately forecast the future demographic-political path, which amplifies the temporary baby-boom price response. When forecasts place more weight on that future path, the price hump and the boom cohort’s net-

worth swing are dampened. The sensitivity checks indicate that the timing of this mechanism is not driven by a particular calibration of non-financial voting preferences or by the main voting weights, although these assumptions can change the level of the response.

The results also speak to current debates over housing-supply reform. If demographic change raises the political weight of households with an interest in preserving housing scarcity, then reforms that expand supply may face stronger opposition precisely when affordability pressures are rising. Policies that reduce the private capital losses from new construction, or that broaden participation in local housing decisions, may therefore be important complements to planning reform.

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A Numerical Algorithm

The numerical solution has two parts. The first solves the stationary political-economic accounting equilibrium for a fixed demographic structure. The second solves transition paths under a specified rule for expectations.

A.1 Stationary Equilibrium

For a fixed demographic structure, the numerical solution follows these steps.

1. The household's value functions are solved recursively for a fixed value of p^h , starting in the final period of life and working backwards. This provides solutions for the household value functions $\{V_j^{rent}, V_j^{adjust}, V_j^{non-adjust}\}$ and the resulting decision rules for consumption, housing and liquid assets as a function of the households' idiosyncratic state variables.
2. The rental price of housing is pinned down by the cost of housing p^h .
3. We then calculate how these value functions change if housing supply is marginally restricted. Operationally, this is evaluated through the small persistent price change ϵ^p implied by lower supply. The difference in value functions is used to calculate preferences over housing supply across households, Θ .
4. Next we solve for the density of households across the life cycle using these decision rules to solve for the distribution of households across the idiosyncratic state variables.
5. We then use these solutions to calculate the share of households voting to restrict housing supply, $S(p^h)$. We solve for the root of $S(p^h) - 0.5 = 0$ using a bracketed price search. This avoids relying on a directional update rule and directly implements the median-voter condition.
6. Given fixed age weights and household formation rates, we iterate these decision rules until the cross-sectional distribution is stationary under the imposed demographic structure.

A.2 Transition Paths with Forecast Rules

The main text defines the forecast path and the update rule in equations 9 and 22. The numerical transition path calculation implements these objects as follows.

1. Fix the demographic path $\Omega = \{\Omega_t\}_{t=0}^T$ and an initial collection of perceived continuation price paths. For the current-price case, future prices are set equal to the current house price at date t .
2. Given the perceived continuation prices, solve the household problem backwards along the transition using the same tenure, saving and housing choice problem as in the stationary case.

3. Simulate the economy forward using the resulting decision rules. In each period, aggregate the voting rule and determine the housing-supply restriction. Conditional on the permitted housing stock, compute the market-clearing house price. This gives the generated price path $\mathcal{T}(\hat{\mathcal{P}}; \Omega)$.
4. For $\lambda > 0$, update the perceived continuation paths toward the generated path using equation 22. The reported transition exercises use a finite number of forecast revisions and report the associated revision counts and forecast discrepancies in the numerical output.